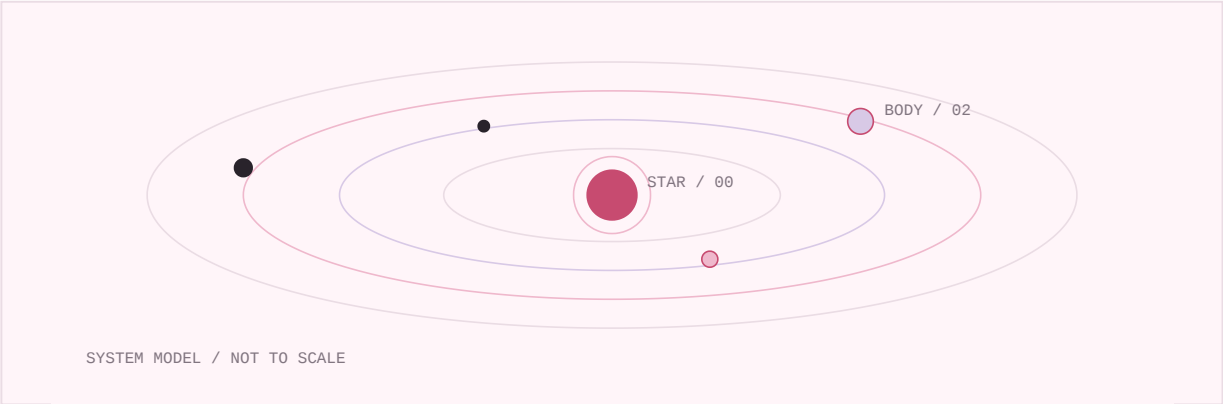
PLANETARY CARTOGRAPHY / PROJECT DOCUMENT 001

# ALL PLANETERAC

惑星地図 / An interactive planetary system atlas




| 01 | 02 | 03 |
| --- | --- | --- |
| MAP | INSPECT | ARCHIVE |

## PROJECT PROFILE

ALL PLANETERAC is a browser-based planetary cartography and world-building project. It brings together a visual orbital map, editable celestial-body records, a compact system overview, and portable data export in one interface.

This field guide documents the project's concepts, orbital model, body records, and data workflow.

> **DESIGN STATEMENT**
>
> A small atlas for imagined worlds — visual enough to explore, structured enough to study, and portable enough to preserve.

AXAYACATL / INTERACTIVE SYSTEMS

# One system.
# Multiple views.

The project treats a planetary system as both a visual map and a structured dataset. The map supports spatial reading; the inspector supports close inspection; the archive tools preserve and move a system between sessions.

| Layer | Purpose | Working question |
|---|---|---|
| Orbital map | Shows the star, orbital paths, and body positions. | Where does each body sit in the system? |
| Body inspector | Edits a selected body's descriptive and numeric fields. | What defines this body? |
| System summary | Collects high-level counts and system information. | What is the overall structure? |
| Body list | Provides direct selection and record management. | Which object should I inspect? |
| Archive tools | Support local persistence and JSON transfer. | How can this system be preserved? |

## DESIGN PRINCIPLES

- **Readable first.** Labels and controls remain legible without collisions, even in narrow viewports.
- **Spatial consistency.** A body's position corresponds to its assigned orbit, not its order in a list.
- **Inspectable records.** Numeric fields and notes can be edited without losing visual context.
- **Portable data.** Export and import make a system reusable outside the current browser session.
- **Quiet visual language.** Pale rose, white space, restrained borders, and technical typography keep attention on the model.

## PROJECT VOCABULARY

> **Body** — a planet, moon, dwarf planet, or other modeled object.
> **Orbit** — the body's assigned path around the system center.
> **Phase** — the body's angular position along that path.
> **System file** — the portable JSON representation of a system.

## VISUAL LANGUAGE

Rose marks key actions and orbital emphasis. Lilac provides a secondary object color. Fine borders separate information without dominating it. Monospaced labels create a cartographic, instrument-panel character.

# Map the motion.

The orbital map should communicate both structure and motion. Each body should have its own orbital radius and a stable phase value so that it does not jump to a new position when the list order changes.

## POSITION MODEL

```
x = cx + r · cos(θ)
y = cy + r · sin(θ)
```

Here, **cx** and **cy** are the map center, **r** is the display-scaled orbital radius, and **θ** is the body's current orbital phase. A model can advance θ over time to animate the system.

Display coordinates are a visualization, not a promise of physically exact astronomical scale. If the interface uses AU values, clearly distinguish the physical value from the display scaling.

## RECOMMENDED INTERACTION MODEL

**01** **Select**

Click a body marker or its entry in the body list to open the matching inspector.

**02** **Inspect**

Read the selected body's orbit, size, temperature, moon count, notes, and color.

**03** **Navigate**

Use zoom and reset controls to examine the map without losing the selected body.

**04** **Animate**

Play or pause orbital motion; make speed controls explicit and understandable.

**05** **Edit**

Change a record and reflect the update in the map and system summary.

## VISUAL LEGIBILITY CHECKLIST

- Orbit rings share a consistent center and correspond to their assigned bodies.
- The selected orbit and body are visually distinct from unselected objects.
- Labels have enough separation to avoid collisions; long names can wrap or use callouts.
- The central star remains visible and is not covered by another marker.
- Small screens retain usable controls and a readable inspector.
- Animation can be paused and respects reduced-motion preferences.

# Objects as data.

A body record combines measurable attributes with world-building context. Keep units explicit and use the same field meanings throughout the interface and exported data.

| Field | Meaning | Validation / interface note |
| --- | --- | --- |
| `name` | Display name for the object. | Required; render safely as text, not executable markup. |
| `classification` | Object type or category. | Use clear category labels; support custom terms if intended. |
| `orbit` | Assigned orbital radius, usually AU or a project-defined unit. | Numeric and non-negative; display the unit. |
| `size` | Body size or radius value. | State whether this is a relative display size or a physical measurement. |
| `temperature` | Estimated surface or representative temperature. | Include °C or K; avoid implying false precision. |
| `moons` | Moon count associated with the body. | Whole number of zero or greater. |
| `notes` | Free-form description and world-building notes. | Wrap long text and preserve intended line breaks. |
| `color` | Visual color used to identify the body. | Provide visible selection and sufficient contrast. |
| `phase / id` | Stable identity and initial angular position. | Keep identity stable through editing, sorting, and import/export. |

### DATA QUALITY RULES

- Reject or normalize empty names and invalid numeric values before saving.
- Use stable IDs so selection and orbital phase survive sorting.
- Keep visible field labels aligned with exported data keys.
- Preserve unknown extra fields when practical, or warn before discarding them.
- When a body is deleted, update the map, list, selected inspector, and summary together.

> **WORLD-BUILDING NOTE**
>
> The record can describe a physically plausible planet, an invented celestial object, or a speculative world. Distinguish observed values, calculated estimates, and fictional parameters in the interface when that distinction matters.

# Preserve the system.

The archive layer should make experimentation safe. Local saving is convenient for the current browser; JSON export provides an explicit backup that can be moved, versioned, or inspected.

## RECOMMENDED WORKFLOW

**01** **BUILD**

Create or edit the system and add bodies.

**02** **REVIEW**

Check orbit values, labels, units, and the map.

**03** **SAVE**

Use local save to retain the current browser state.

**04** **EXPORT**

Download a JSON snapshot before major changes.

**05** **RESTORE**

Import a known-good JSON file if the working state needs recovery.

## EXAMPLE PORTABLE RECORD

```
{
  "name": "Example System",
  "bodies": [
    {
      "id": "body-001",
      "name": "Example I",
      "classification": "terrestrial",
      "orbit": 1.0,
      "size": 1.0,
      "temperature": 15,
      "moons": 1,
      "notes": "Sample record",
      "color": "#efb8cc",
      "phase": 1.2
    }
  ]
}
```

Illustrative schema only. Match field names to the actual HTML implementation when connecting import and export.

## TROUBLESHOOTING

| Symptom | Check |
|---|---|
| Body jumps after sorting | Store stable phase and identity on the body rather than deriving angle from list position. |
| Labels overlap | Add label offsets, collision-aware placement, shorter labels, or selected-body callouts. |
| Import looks incomplete | Validate JSON shape and required fields; report invalid records clearly. |

| Symptom | Check |
| --- | --- |
| State disappears | Check browser storage availability and provide an export/backup path. |
| Artwork does not load | Check URL, filename capitalization, network access, and remote-host availability; provide a fallback. |

# Ready for the next orbit.

A planetary system editor is most convincing when every visual element reflects the underlying record and every record can be recovered. The map is the interface; the data model is the source of truth.

### RELEASE CHECKLIST

- ☐ Orbital positions map to each body's assigned orbit and stable phase.
- ☐ Play/pause and navigation controls have clear, immediate effects.
- ☐ Selecting a marker and selecting a list entry identify the same body.
- ☐ The inspector remains readable at desktop and mobile widths.
- ☐ Creating, editing, and deleting a body updates every dependent view.
- ☐ Import and export handle malformed data without corrupting the active system.
- ☐ Remote artwork has a fallback and is clearly credited where appropriate.
- ☐ The system can be saved and restored using a documented path.

PROJECT FIELD GUIDE / 2026

# ALL PLANETERAC

A small atlas for imagined worlds.

### DOCUMENT PURPOSE

Companion documentation for the ALL PLANETERAC interactive planetary cartography project.

### IMPLEMENTATION NOTE

Orbital equations and example data are explanatory references. Confirm field names and features against the current HTML build.